

1. If repetitions are not allowed, how many 7-digit numbers can be formed from the digits 1, 2, 3, 4, 5, 6, 7?

Using the permutation formula:

$P(n, r) = \frac{n!}{(n-r)!}$, where $P(n, r)$ is the possible number of 7-digit numbers that can be formed from the digits 1-7, n is the total number of digits that being used (7), and r is the required number/length of the digit to be formed (7)

$$\therefore P(7, 7) = \frac{7!}{(7-7)!} = \frac{7!}{0!} = 7! = 5040$$

$\therefore$ There are 5040 different 7-digit numbers that can be formed from the digits 1-7. ■

2. The number of ways in which p like jobs may be assigned to $(q+1)$ persons, if no person is left unemployed and we have $p > q$.

Following the binomial theorem:

$$(q+1)^p = \sum_{j=0}^p \binom{p}{j} q^{p-j} \cdot 1^j = \sum_{j=0}^p \binom{p}{j} q^{p-j}$$

$\therefore$ There are $(q+1)^p - 1$ possibilities. ■

3. 3 pigeon holes marked A, B, C. In how many ways can I arrange 10 different postcards so that 5 of them are in A, 3 in B and 2 in C?

If 10 objects are placed in 3 pigeon holes with 5 in A there the possibilities for A include

$$C(n, r) = \frac{n!}{(n-r)!r!} = C(10, 5) = \frac{10!}{5!5!} = 252$$

For the potential 3 postcards in B:

$$C(n, r) = \frac{n!}{(n-r)!r!} = C(10, 3) = \frac{10!}{7!3!} = 120$$

For the potential 2 postcards in C:

$$C(n, r) = \frac{n!}{(n-r)!r!} = C(10, 2) = \frac{10!}{8!2!} = 45$$

The total number of ways to arrange the 10 different postcards is All the possibilities for A, B, and C added together.

$$\text{Total} = A + B + C = 252 + 120 + 45 = 417$$

$\therefore$ 417 ways to arrange the 10 different postcards so that 5 of them are in A, 3 in B and 2 in C. ■

4. Consider all bit strings of length 14. How many have exactly eleven 1's such that none of these 1's are adjacent to each other?

For the bit string to have all 11 of its 1's not being adjacent to each other it would require at least 21 spaces.

$\therefore$ There are 0 possible 14 length bit strings that have exactly 11 1's such that non of the 1's are adjacent to each other. ■